Bellabeat Fitness Data Analysis – Pandas

This analysis explores user activity patterns using the Bellabeat fitness dataset. The objective is to understand daily movement trends, sedentary behavior, activity intensity distribution, and relationships between steps and calorie expenditure.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

1. Data Loading

The dataset is loaded into a Pandas DataFrame for analysis.

```
df = pd.read_csv("/dailyActivity_merged.csv")
display(df.head())
```

	Id	ActivityDate	TotalSteps	TotalDistance	TrackerDistance	LoggedActivitiesDistance	VeryActiveDistance	Moderatel
0	1503960366	4/12/2016	13162	8.50	8.50	0.0	1.88	
1	1503960366	4/13/2016	10735	6.97	6.97	0.0	1.57	
2	1503960366	4/14/2016	10460	6.74	6.74	0.0	2.44	
3	1503960366	4/15/2016	9762	6.28	6.28	0.0	2.14	
4	1503960366	4/16/2016	12669	8.16	8.16	0.0	2.71	

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 940 entries, 0 to 939
Data columns (total 15 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Id                        940 non-null    int64
 1   ActivityDate              940 non-null    object
 2   TotalSteps                940 non-null    int64
 3   TotalDistance             940 non-null    float64
 4   TrackerDistance           940 non-null    float64
 5   LoggedActivitiesDistance  940 non-null    float64
 6   VeryActiveDistance        940 non-null    float64
 7   ModeratelyActiveDistance  940 non-null    float64
 8   LightActiveDistance       940 non-null    float64
 9   SedentaryActiveDistance   940 non-null    float64
 10  VeryActiveMinutes         940 non-null    int64
 11  FairlyActiveMinutes       940 non-null    int64
 12  LightlyActiveMinutes      940 non-null    int64
 13  SedentaryMinutes          940 non-null    int64
 14  Calories                  940 non-null    int64
dtypes: float64(7), int64(7), object(1)
memory usage: 110.3+ KB
```

The dataset contains 940 daily records and 15 columns. All columns contain non-null values, indicating no missing data issues.

The dataset includes information such as:

- Total Steps
- Calories Burned
- Activity Minutes (Light, Fairly, Very Active)
- Sedentary Minutes

Data Understanding

This section explores the structure of the dataset and provides an overview of the available variables including steps, activity minutes, distance, and calories burned.

```
df.shape
```

```
(940, 15)
```

```
df.info()
```

```
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RangeIndex: 940 entries, 0 to 939
Data columns (total 15 columns):
 #   Column                    Non-Null Count  Dtype
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 8   LightActiveDistance       940 non-null    float64
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 13  SedentaryMinutes          940 non-null    int64
 14  Calories                  940 non-null    int64
dtypes: float64(7), int64(7), object(1)
memory usage: 110.3+ KB
```

```
df.isnull().sum()
```

	0
Id	0
ActivityDate	0
TotalSteps	0
TotalDistance	0
TrackerDistance	0
LoggedActivitiesDistance	0
VeryActiveDistance	0
ModeratelyActiveDistance	0
LightActiveDistance	0
SedentaryActiveDistance	0
VeryActiveMinutes	0
FairlyActiveMinutes	0
LightlyActiveMinutes	0
SedentaryMinutes	0
Calories	0

dtype: int64

The dataset contains no missing values, indicating the data is complete for analysis.

```
df.duplicated().sum()
```

```
np.int64(0)
```

No duplicate records were found in the dataset.

2. Data Preparation

The ActivityDate column is converted into datetime format. A new column named "Weekday" is created to analyze weekly activity patterns.

```
df['ActivityDate'] = pd.to_datetime(df['ActivityDate'])
df['Weekday'] = df['ActivityDate'].dt.day_name()

df.head()
```

	Id	ActivityDate	TotalSteps	TotalDistance	TrackerDistance	LoggedActivitiesDistance	VeryActiveDistance	Moderatel
0	1503960366	2016-04-12	13162	8.50	8.50	0.0	1.88	
1	1503960366	2016-04-13	10735	6.97	6.97	0.0	1.57	
2	1503960366	2016-04-14	10460	6.74	6.74	0.0	2.44	
3	1503960366	2016-04-15	9762	6.28	6.28	0.0	2.14	
4	1503960366	2016-04-16	12669	8.16	8.16	0.0	2.71	

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Feature Engineering

New variables are created to enhance analysis, including weekday extraction and total active minutes.

```
df['Weekday'] = df['ActivityDate'].dt.day_name()

df['TotalActiveMinutes'] = (
    df['VeryActiveMinutes']
    + df['FairlyActiveMinutes']
    + df['LightlyActiveMinutes']
)
```

3. Weekly Activity Pattern

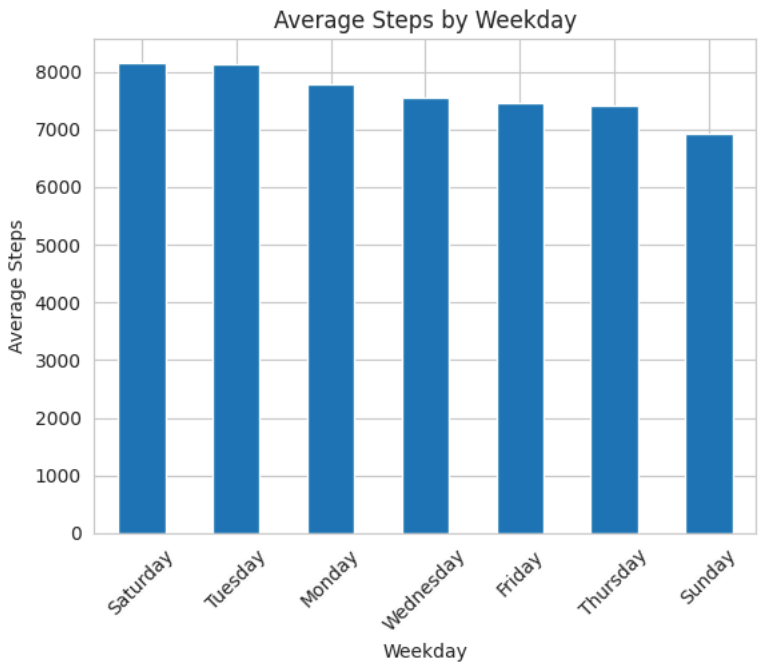
```
weekday_steps = df.groupby('Weekday')['TotalSteps'].mean().sort_values(ascending=False)

weekday_steps
```

Weekday	TotalSteps
Saturday	8152.975806
Tuesday	8125.006579
Monday	7780.866667
Wednesday	7559.373333
Friday	7448.230159
Thursday	7405.836735
Sunday	6933.231405

dtype: float64

```
plt.figure()
weekday_steps.plot(kind='bar')
plt.title("Average Steps by Weekday")
plt.ylabel("Average Steps")
plt.xticks(rotation=45)
plt.show()
```



The bar chart shows that Saturday records the highest average steps, while Sunday records the lowest.

This indicates increased engagement at the beginning of the weekend and reduced activity at the end.

4. Sedentary vs Active Behavior

```
df['TotalActiveMinutes'] = (
    df['LightlyActiveMinutes'] +
    df['FairlyActiveMinutes'] +
    df['VeryActiveMinutes']
)

df[['SedentaryMinutes','TotalActiveMinutes']].mean()
```

	0
SedentaryMinutes	991.210638
TotalActiveMinutes	227.542553

dtype: float64

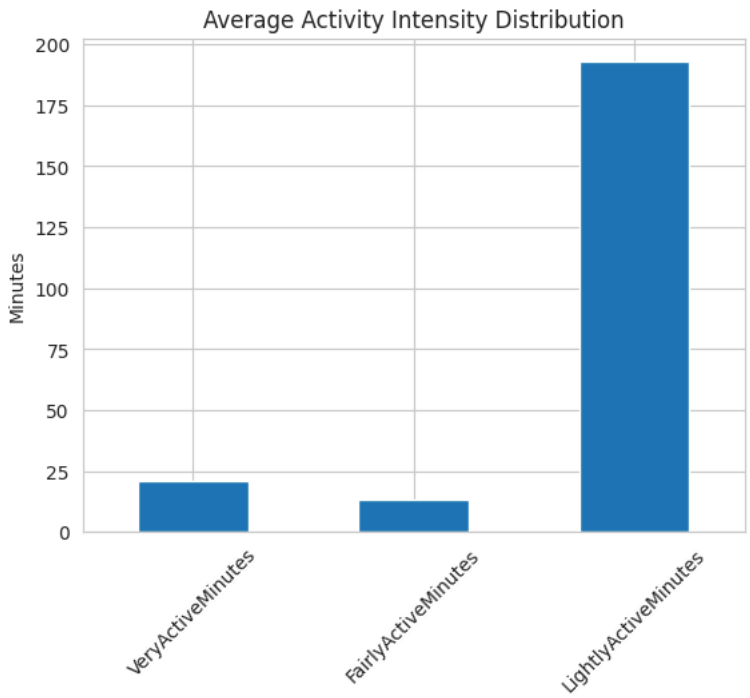
On average, users spend approximately 991 minutes per day sedentary, compared to only 227 active minutes.

This suggests that nearly 70% of daily time is inactive, indicating a predominantly sedentary lifestyle.

5. Activity Intensity Distribution

```
intensity_avg = df[['VeryActiveMinutes','FairlyActiveMinutes','LightlyActiveMinutes']].mean()

plt.figure()
intensity_avg.plot(kind='bar')
plt.title("Average Activity Intensity Distribution")
plt.ylabel("Minutes")
plt.xticks(rotation=45)
plt.show()
```



The majority of activity falls under light intensity.

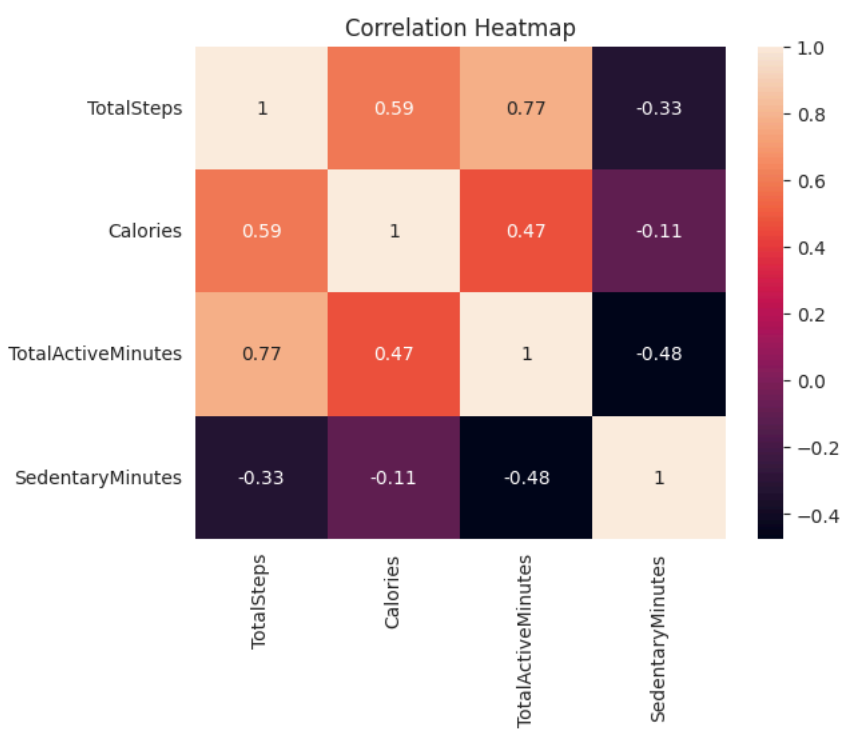
Approximately 85% of total active time consists of light activity, while very active minutes contribute a small portion.

This indicates that users engage more in casual movement rather than structured or high-intensity workouts.

6. Correlation Analysis

```
correlation = df[['TotalSteps','Calories','TotalActiveMinutes','SedentaryMinutes']].corr()

plt.figure()
sns.heatmap(correlation, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



The heatmap reveals:

- A strong positive correlation (0.77) between total steps and active minutes.
- A moderate positive correlation (0.59) between steps and calories burned.
- A negative correlation between active minutes and sedentary minutes.

This confirms that increased movement significantly impacts calorie expenditure.

7. Activity Level Segmentation

High activity users burn significantly more calories than low activity users. The difference in calorie expenditure between low and high activity groups demonstrates the strong impact of movement intensity on energy expenditure.

```
def get_activity_level(total_steps):
    if total_steps < 5000:
        return 'Low Activity'
    elif 5000 <= total_steps < 10000:
        return 'Moderate Activity'
    else:
        return 'High Activity'

df['ActivityLevel'] = df['TotalSteps'].apply(get_activity_level)

segmentation = df.groupby('ActivityLevel')[['TotalSteps','Calories']].mean().round(2)
segmentation
```

ActivityLevel	TotalSteps	Calories
High Activity	13337.40	2743.58
Low Activity	2128.17	1806.81
Moderate Activity	7465.78	2355.16

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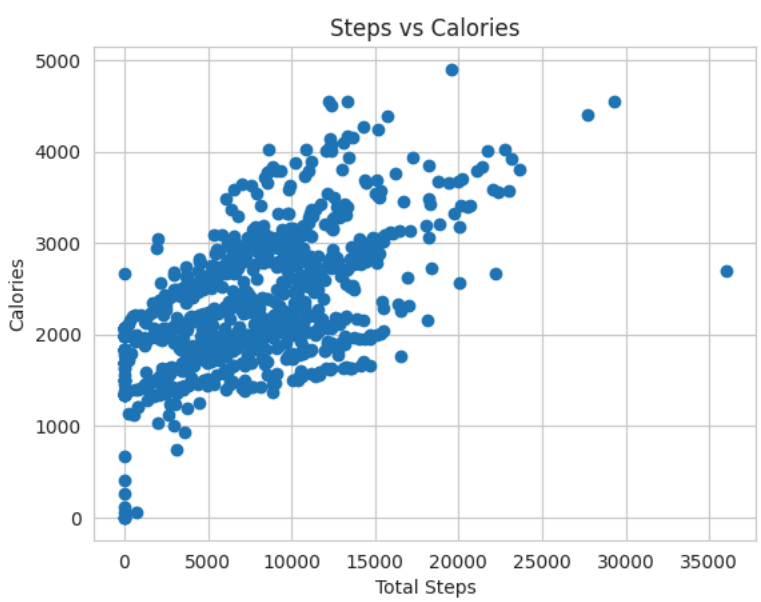
8. Steps vs Calories Relationship

```
plt.figure()
plt.scatter(df['TotalSteps'], df['Calories'])
plt.title("Steps vs Calories")
plt.xlabel("Total Steps")
plt.ylabel("Calories")
plt.show()
```



The analysis reveals that while users engage in daily movement, most activity remains light intensity, and sedentary time is significantly high.

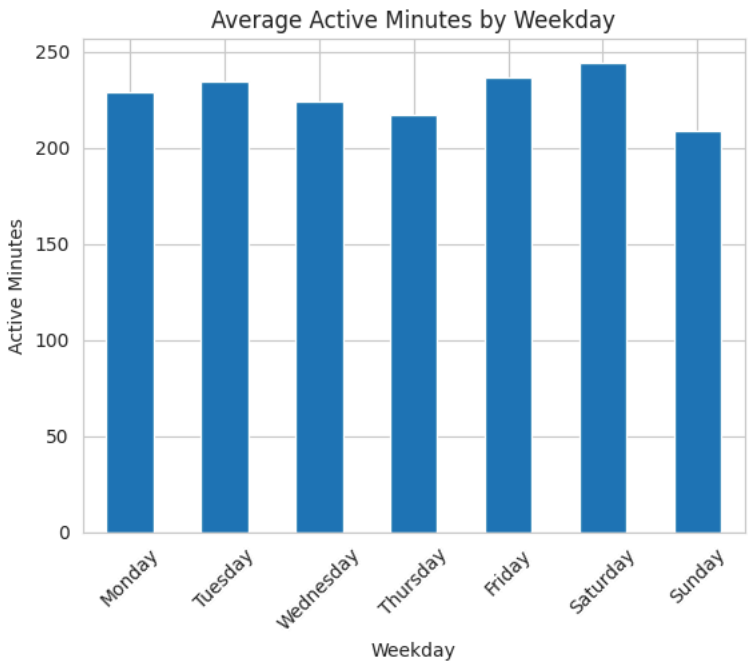
There is a clear positive relationship between steps and calorie expenditure. Increasing movement levels can significantly improve energy burn and overall engagement.

These findings support the need for strategies that encourage higher activity intensity and reduce sedentary behavior.

9. Average Active Minutes by Weekday

```
weekday_active = df.groupby('Weekday')['TotalActiveMinutes'].mean().reindex(
    ["Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"]
)

plt.figure()
weekday_active.plot(kind='bar')
plt.title("Average Active Minutes by Weekday")
plt.ylabel("Active Minutes")
plt.xticks(rotation=45)
plt.show()
```



10. Business Recommendations

Based on the analysis:

- Target low activity users to increase engagement.
- Introduce Sunday motivation campaigns to reduce inactivity.
- Promote intensity-based challenges.
- Offer premium analytics features for high-activity users.